How Valuable Is Financial Flexibility when Revenue Stops? Evidence from the COVID-19 Crisis

Replication Results

Original-vintage sample

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Advanced Financial Theorem

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Appendix A
Sample selection procedure

Original-vintage sample. Compustat North America firms with fiscal year 2019 data, after each screen.

Number of firms with fiscal year 2019 data	11,612
Not available by Mar 12, 2020 (later 10-K or fiscal year ending after Dec 2019)	−8,355
Number of firms with fiscal year 2019 data available in March 2020	3,258
Duplicate data	0
SIC codes 6000-6999 (financial firms)	−826
SIC codes 4900-4949 (utility firms)	−183
SIC codes 8000s and 9000s (not-for-profit and governmental firms)	−137
Non-U.S. headquarters	−121
Missing cash and short-term investments	−4
Missing stock price data in Security Daily	−53
An average share price $<$ \$1	−87
Security type not equal to “common, ordinary”	−4
Number of firms after all screens	1,843

Table C1
Summary statistics

Summary statistics for the raw data and the derived measures of COVID-19 exposure of firms and industries. TQ (BQ) indicates an indicator for the top (bottom) quartile.

	N	P25	Median	Mean	P75	SD
Affected share	1,826	9.000	10.000	14.461	14.000	12.317
Teamwork share	1,826	17.000	25.000	24.789	25.000	10.654
Customer share	1,826	7.000	8.000	13.587	17.000	12.938
Presence share	1,826	3.000	9.000	11.154	11.000	12.934
Q2 2020 year-on-year change in sales	1,697	−0.337	−0.111	0.261	0.077	6.678
Share of jobs that can be performed from home	1,837	0.210	0.278	0.375	0.506	0.231
Labor intensity as number of employees / sales	1,738	0.002	0.003	0.019	0.005	0.279
Affected share (TQ)	1,826	0.000	0.000	0.244	0.000	0.430
Teamwork share (TQ)	1,826	0.000	0.000	0.228	0.000	0.420
Customer share (TQ)	1,826	0.000	0.000	0.214	0.000	0.410
Presence share (TQ)	1,826	0.000	0.000	0.208	0.000	0.406
Q2 2020 year-on-year change in sales (BQ)	1,697	0.000	0.000	0.250	1.000	0.433
Share of jobs that can be performed from home (BQ)	1,837	0.000	0.000	0.250	1.000	0.433
Labor intensity based on number of employees (TQ)	1,738	0.000	0.000	0.250	0.750	0.433
Manually classified as highly affected	1,826	0.000	0.000	0.245	0.000	0.430

Table C2
Correlation matrix

Correlations among the measures of COVID-19 exposure of firms and industries.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(1) Affected share (TQ)	1.00							
(2) Teamwork share (TQ)	0.18	1.00						
(3) Customer share (TQ)	0.53	0.24	1.00					
(4) Presence share (TQ)	0.39	0.13	−0.07	1.00				
(5) Q2 2020 year-on-year change in sales (BQ)	0.19	0.05	0.04	0.11	1.00			
(6) Share of jobs that can be performed from home (BQ)	0.17	−0.18	0.08	0.26	0.13	1.00		
(7) Labor intensity based on number of employees (TQ)	−0.01	−0.07	0.11	−0.18	0.02	0.00	1.00	
(8) Manually classified as highly affected	0.97	0.21	0.50	0.43	0.23	0.17	−0.01	1

Table 1
Summary statistics

Summary statistics for the dependent and independent variables. The sample consists of all nonfinancial and nonutility U.S. firms with fiscal year 2019 data in Compustat that pass the screens in Appendix A. GFC, global financial crisis.

	N	Min	P1	P5	P25	Median	Mean	P75	P95	P99	Max	SD
Dependent variables: COVID-19-related												
Cum. daily exc. returns (Feb 3 - Mar 23, 2020)	1,843	−0.910	−0.825	−0.707	−0.513	−0.380	−0.373	−0.259	−0.032	0.315	3.456	0.243
Daily exc. returns (March 24, 2020)	1,834	−0.481	−0.097	−0.019	0.045	0.091	0.097	0.139	0.245	0.370	0.597	0.087
ROA (cumulative Q1 + Q2 2020)	1,783	−0.807	−0.806	−0.333	−0.031	0.031	−0.017	0.061	0.121	0.215	0.215	0.156
Dependent variables: GFC-related												
Cum. daily exc. returns (Sep 15 - Dec 31, 2008)	3,179	−0.998	−0.939	−0.824	−0.589	−0.407	−0.393	−0.231	0.031	0.385	11.662	0.368
Cum. daily exc. returns (Sep 15, 2008 - Mar 9, 2009)	3,179	−0.999	−0.974	−0.904	−0.728	−0.551	−0.510	−0.370	−0.001	0.475	9.328	0.370
Independent and other variables												
Assets ($m)	1,843	0.002	7.615	28.460	284.544	1,240.810	8,062.455	4,433.367	33,726.300	135,306.420	551,669.000	29,287.936
Cash / assets	1,843	0.000	0.001	0.006	0.038	0.116	0.239	0.352	0.864	0.961	0.962	0.274
St debt / assets	1,843	0.000	0.000	0.000	0.005	0.013	0.031	0.034	0.119	0.353	0.355	0.054
nSt debt / assets	1,843	−0.958	−0.958	−0.854	−0.325	−0.087	−0.207	−0.011	0.049	0.293	0.295	0.285
Lt debt / assets	1,838	0.000	0.000	0.000	0.072	0.244	0.271	0.403	0.686	1.112	1.116	0.232
Book debt / assets	1,838	0.000	0.000	0.000	0.102	0.280	0.306	0.442	0.749	1.215	1.228	0.247
Payout / assets	1,843	0.000	0.000	0.000	0.000	0.011	0.035	0.040	0.160	0.359	0.363	0.061
Ind: 3-year cum. payout / assets top quartile	1,816	0.000	0.000	0.000	0.000	0.000	0.250	0.250	1.000	1.000	1.000	0.433
Ind: COVID-19 conglomerate	1,843	0.000	0.000	0.000	0.000	0.000	0.088	0.000	1.000	1.000	1.000	0.283
Ind: IG debt rating	1,843	0.000	0.000	0.000	0.000	0.000	0.135	0.000	1.000	1.000	1.000	0.342
Ind: HY debt rating	1,843	0.000	0.000	0.000	0.000	0.000	0.184	0.000	1.000	1.000	1.000	0.388
RD / assets	1,843	0.000	0.000	0.000	0.000	0.013	0.094	0.096	0.447	0.828	5.576	0.226
SGA / sales	1,569	0.018	0.018	0.042	0.113	0.222	0.438	0.426	1.235	6.924	7.055	0.874
Capex / lagged assets	1,839	0.000	0.000	0.002	0.013	0.028	0.045	0.056	0.161	0.271	0.273	0.051
COGS / sales	1,749	0.000	0.058	0.160	0.414	0.626	0.601	0.798	1.000	1.000	1.000	0.251
KZ index	1,797	−4,240.199	−592.610	−112.165	−12.098	−2.434	−31.300	0.748	3.442	30.871	2,944.772	232.770
FC (KZ index)	1,797	0.000	0.000	0.000	0.000	0.000	0.250	0.000	1.000	1.000	1.000	0.433
WW index	1,733	−1.631	−0.551	−0.494	−0.394	−0.297	−0.140	−0.144	0.121	7.348	7.561	0.967
FC (WW index)	1,733	0.000	0.000	0.000	0.000	0.000	0.250	0.000	1.000	1.000	1.000	0.433
SA index	1,843	−4.637	−4.637	−4.637	−4.237	−3.638	−3.611	−3.154	−2.346	−1.484	6.550	0.802
FC (SA index)	1,843	0.000	0.000	0.000	0.000	0.000	0.250	0.500	1.000	1.000	1.000	0.433
HR cash ex ante	1,839	−20,905.000	−1,825.666	−229.863	9.160	114.761	1,034.631	448.811	3,452.000	14,683.540	211,611.000	7,990.135
FC (HR cash ex ante)	1,839	0.000	0.000	0.000	0.000	0.000	0.190	0.000	1.000	1.000	1.000	0.393
Equity beta	1,832	−3.184	−0.239	0.245	0.803	1.122	1.124	1.468	2.011	2.391	4.227	0.553
BM at EoY 2019	1,839	−1.018	−0.983	−0.045	0.149	0.325	0.471	0.650	1.570	2.976	2.985	0.573
ln(MVE) at EoY 2019	1,843	1.367	2.097	3.708	5.816	7.268	7.185	8.579	10.785	12.132	12.142	2.092
Momentum 2019	1,842	−0.969	−0.835	−0.564	−0.083	0.187	0.234	0.461	1.072	2.752	2.753	0.557
Profitability	1,843	−6.900	−0.880	−0.330	0.135	0.246	0.233	0.376	0.674	1.136	1.138	0.356

Table 2
Correlation matrix

Correlations among the dependent and independent variables for the sample described in Table 1.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)	(25)
(1) Cum. daily exc. returns (Feb 3 - Mar 23, 2020)	1.00																								
(2) Daily exc. returns (March 24, 2020)	−0.38	1.00																							
(3) ROA (cumulative Q1 + Q2 2020)	0.00	0.13	1.00																						
(4) Assets ($m)	0.01	0.04	0.11	1.00																					
(5) Cash / assets	0.21	−0.11	−0.59	−0.13	1.00																				
(6) St debt / assets	−0.02	0.00	−0.07	0.06	−0.11	1.00																			
(7) nSt debt / assets	−0.21	0.10	0.53	0.14	−0.97	0.32	1.00																		
(8) Lt debt / assets	−0.20	0.16	0.22	0.05	−0.32	0.01	0.31	1.00																	
(9) Book debt / assets	−0.19	0.14	0.15	0.05	−0.32	0.31	0.38	0.94	1.00																
(10) Payout / assets	0.02	0.05	0.26	0.10	−0.12	0.03	0.12	0.10	0.11	1.00															
(11) Ind: 3-year cum. payout / assets top quartile	0.02	0.02	0.24	0.12	−0.14	0.01	0.14	0.07	0.06	0.67	1.00														
(12) Ind: COVID-19 conglomerate	−0.02	−0.01	0.11	0.08	−0.14	0.03	0.14	0.04	0.04	0.02	0.06	1.00													
(13) Ind: IG debt rating	0.00	0.07	0.18	0.44	−0.21	0.06	0.21	0.06	0.06	0.22	0.29	0.08	1.00												
(14) Ind: HY debt rating	−0.20	0.14	0.16	−0.02	−0.28	0.02	0.27	0.33	0.31	−0.06	−0.07	0.02	−0.19	1.00											
(15) RD / assets	0.17	−0.13	−0.62	−0.08	0.53	0.03	−0.50	−0.15	−0.12	−0.10	−0.10	−0.10	−0.12	−0.17	1.00										
(16) SGA / sales	0.05	−0.03	−0.58	−0.07	0.54	−0.01	−0.50	−0.12	−0.11	−0.09	−0.10	−0.09	−0.11	−0.15	0.60	1.00									
(17) Capex / lagged assets	−0.14	0.09	0.09	0.02	−0.23	−0.01	0.22	0.08	0.07	0.00	0.00	0.04	0.00	0.05	−0.18	−0.11	1.00								
(18) COGS / sales	−0.03	−0.03	−0.20	−0.05	0.02	0.09	0.00	−0.05	−0.02	−0.16	−0.11	0.09	−0.05	0.06	0.12	−0.25	−0.05	1.00							
(19) FC (KZ index)	−0.14	0.08	−0.11	−0.05	−0.22	0.14	0.25	0.34	0.37	−0.20	−0.21	−0.01	−0.13	0.21	0.01	0.02	0.25	0.11	1.00						
(20) FC (WW index)	0.18	−0.14	−0.45	−0.14	0.57	−0.01	−0.54	−0.17	−0.15	−0.16	−0.15	−0.13	−0.19	−0.19	0.43	0.38	−0.18	0.03	−0.04	1.00					
(21) FC (SA index)	0.14	−0.16	−0.53	−0.15	0.58	0.00	−0.55	−0.28	−0.24	−0.18	−0.17	−0.14	−0.23	−0.26	0.41	0.40	−0.09	0.03	−0.02	0.49	1.00				
(22) FC (HR cash ex ante)	−0.08	−0.06	−0.19	−0.03	−0.16	0.09	0.18	0.12	0.15	−0.05	−0.04	0.03	−0.05	0.00	0.10	0.11	0.23	0.04	0.22	0.00	0.06	1.00			
(23) Equity beta	−0.09	0.15	−0.02	−0.05	0.12	−0.10	−0.15	0.03	−0.02	−0.11	−0.17	−0.08	−0.06	0.09	0.01	−0.02	0.08	0.02	0.05	0.02	−0.07	−0.05	1.00		
(24) BM at EoY	−0.09	−0.06	0.04	−0.02	−0.22	−0.05	0.19	−0.18	−0.20	−0.18	−0.19	0.08	−0.08	0.08	−0.17	−0.10	0.12	0.17	0.14	−0.13	−0.10	0.09	0.09	1.00	
(25) ln(MVE) at EoY	−0.02	0.22	0.39	0.44	−0.25	−0.07	0.22	0.17	0.13	0.30	0.31	0.05	0.54	0.10	−0.25	−0.20	0.05	−0.24	−0.21	−0.35	−0.50	−0.17	0.14	−0.30	1.00
(26) Momentum	0.02	0.10	0.02	0.01	0.09	−0.05	−0.10	−0.05	−0.05	−0.01	−0.02	−0.02	0.03	0.00	−0.02	−0.08	−0.03	−0.02	−0.11	0.00	−0.01	−0.13	−0.02	−0.28	0.25
(27) Profitability	−0.03	0.08	0.53	0.00	−0.35	−0.02	0.32	0.03	0.02	0.24	0.18	0.03	0.05	0.02	−0.64	−0.08	0.07	−0.47	−0.11	−0.24	−0.20	−0.16	−0.08	−0.04	0.16

Table 3

Stock returns, financial flexibility measures, and stock and firm characteristics

Cross-sectional regressions of stock returns in excess of the risk-free rate on firm characteristics. Collapse period: cumulative return from February 3 to March 23, 2020. Stimulus day: return on March 24, 2020. Columns 7 and 8 use the global financial crisis, with cumulative returns from September 15, 2008, to December 31, 2008, and to March 9, 2009, and firm characteristics measured in fiscal year 2006. All regressions include Fama-French 49 industry fixed effects. t-statistics are in parentheses. $*p < .1$; $**p < .05$; $***p < .01$.

	(1) Collapse period	(2) Stimulus day	(3) Collapse period	(4) Stimulus day	(5) Collapse period	(6) Stimulus day	(7) Global financial crisis ending Dec 31, 2008	(8) Global financial crisis ending March 9, 2009
Cash / assets	0.070	0.010			0.071*	0.009	0.123***	0.159***
	(1.64)	(0.64)			(1.68)	(0.55)	(2.87)	(3.87)
St debt / assets	−0.226**	0.084**					−0.208**	−0.310***
	(−2.00)	(1.98)					(−2.07)	(−3.20)
nSt debt / assets			−0.092**	−0.001				
			(−2.46)	(−0.07)				
Lt debt / assets	−0.174***	0.030***	−0.142***	0.019			−0.114***	−0.066
	(−5.85)	(2.70)	(−4.63)	(1.63)			(−2.66)	(−1.60)
Book debt / assets					−0.174***	0.033***		
					(−6.22)	(3.13)		
Ind: IG debt rating	0.016	−0.011	−0.005	−0.003	0.016	−0.010	0.053*	0.051*
	(0.84)	(−1.49)	(−0.25)	(−0.34)	(0.80)	(−1.37)	(1.82)	(1.82)
Ind: HY debt rating			−0.047***	0.016***				
			(−2.89)	(2.70)				
Capex / lagged assets	0.011	0.028	−0.002	0.033	0.009	0.029	−0.018	−0.218***
	(0.09)	(0.58)	(−0.02)	(0.69)	(0.07)	(0.61)	(−0.25)	(−3.12)
SGA / sales	−0.033***	0.009***	−0.034***	0.009***	−0.034***	0.009***	−0.000	0.002
	(−3.56)	(2.66)	(−3.68)	(2.73)	(−3.61)	(2.65)	(−0.06)	(0.39)
COGS / sales	0.028	0.004	0.035	0.002	0.027	0.006	−0.000	−0.015
	(0.78)	(0.31)	(0.98)	(0.18)	(0.74)	(0.44)	(−0.01)	(−0.35)
RD / assets	0.030	−0.017	0.010	−0.006	0.039	−0.015	0.012	0.233***
	(0.32)	(−0.49)	(0.11)	(−0.17)	(0.42)	(−0.43)	(0.19)	(3.73)
Payout / assets	0.104	−0.034	0.069	−0.019	0.109	−0.032	0.122	0.182*
	(1.01)	(−0.87)	(0.67)	(−0.48)	(1.05)	(−0.83)	(1.18)	(1.82)
Equity beta	−0.057***	0.025***	−0.057***	0.025***	−0.059***	0.025***	−0.014	−0.053***
	(−4.65)	(5.34)	(−4.64)	(5.30)	(−4.79)	(5.37)	(−1.16)	(−4.53)
BM at EoY 2019	0.009	−0.006	0.014	−0.008*	0.008	−0.006	−0.032	−0.066***
	(0.72)	(−1.24)	(1.17)	(−1.77)	(0.65)	(−1.23)	(−1.43)	(−3.09)
ln(MVE) at EoY 2019	0.005	0.008***	0.008*	0.007***	0.004	0.008***	0.014**	0.005
	(1.12)	(5.00)	(1.87)	(4.15)	(1.07)	(4.94)	(2.54)	(0.92)
Momentum 2019	−0.006	0.012**	−0.005	0.012**	−0.006	0.012**	0.012	−0.008
	(−0.42)	(2.50)	(−0.38)	(2.46)	(−0.42)	(2.51)	(0.93)	(−0.64)
Profitability	0.058	0.009	0.058	0.007	0.056	0.009	−0.012	−0.024
	(1.54)	(0.64)	(1.58)	(0.52)	(1.49)	(0.62)	(−0.31)	(−0.66)
Observations	1,554	1,549	1,554	1,549	1,554	1,549	2,766	2,766
Adjusted R^2	0.192	0.128	0.196	0.130	0.193	0.128	0.035	0.080

Table 4

Bottom-20 NAICS three-digit industries cumulative returns and industry characteristics: GFC versus COVID-19 crisis

The 20 NAICS three-digit industries with the lowest equal-weighted cumulative excess return (CR) from September 15, 2008, to March 9, 2009 (left) and from February 3 to March 23, 2020 (right). * marks industries in the bottom 20 in both crises.

Global financial crisis					COVID-19 crisis				
NAICS	Affected share	Lt debt / assets	PPE / assets	CR	NAICS	Affected share	Lt debt / assets	PPE / assets	CR
* 314	10.000	0.245	0.354	−0.765	* 451	82.000	0.404	0.481	−0.724
* 721	38.000	0.378	0.622	−0.743	* 213	31.000	0.274	0.540	−0.662
513		0.357	0.030	−0.734	448	90.000	0.496	0.314	−0.639
* 337	9.000	0.187	0.260	−0.710	* 211	17.000	0.320	0.841	−0.630
* 313	9.000	0.242	0.277	−0.710	486	10.000	0.600	0.606	−0.615
515	17.000	0.338	0.176	−0.701	713	44.000	0.627	0.564	−0.607
711	27.000	0.211	0.553	−0.678	* 236	16.000	0.344	0.039	−0.572
* 213	31.000	0.171	0.534	−0.662	324	12.000	0.306	0.503	−0.554
* 451	82.000	0.074	0.271	−0.659	* 314	10.000	0.337	0.348	−0.554
238	16.000	0.169	0.190	−0.657	441	44.000	0.289	0.332	−0.520
* 211	17.000	0.228	0.701	−0.655	* 323	14.000	0.430	0.227	−0.520
* 236	16.000	0.273	0.026	−0.647	481	48.000	0.336	0.732	−0.514
331	9.000	0.166	0.305	−0.638	327	9.000	0.396	0.355	−0.511
321	9.000	0.209	0.287	−0.630	* 721	38.000	0.554	0.484	−0.501
* 323	14.000	0.204	0.244	−0.625	* 313	9.000	0.251	0.339	−0.500
519	10.000	0.058	0.158	−0.623	722	53.000	0.564	0.558	−0.496
492	14.000	0.154	0.347	−0.621	483	9.000	0.319	0.760	−0.495
421		0.190	0.086	−0.618	512	43.000	0.491	0.607	−0.495
322	10.000	0.263	0.426	−0.612	* 337	9.000	0.225	0.311	−0.488
315	10.000	0.120	0.115	−0.611	532	39.000	0.519	0.676	−0.486
Mean	19.333	0.212	0.298	−0.665	Mean	31.350	0.404	0.481	−0.554
Median	14.000	0.206	0.274	−0.656	Median	24.000	0.370	0.494	−0.520

Table 5

Stock returns, financial flexibility measures interacted with COVID-19 exposure measures, and stock and firm characteristics

Cross-sectional regressions of cumulative excess stock returns from February 3 to March 23, 2020, on financial flexibility measures interacted with an indicator for firms highly exposed to COVID-19 under each exposure measure. Controls include equity beta, the 2019 stock return, the book-to-market ratio, the log market value of equity, gross profitability, capex, SG&A, R&D, payouts, and an investment-grade rating indicator. All regressions include Fama-French 49 industry fixed effects. t-statistics are in parentheses. $^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$.

	(1) Affected share (TQ)	(2) Teamwork share (TQ)	(3) Customer share (TQ)	(4) Presence share (TQ)	(5) Q2 2020 year-on-year change in sales (BQ)	(6) Share of jobs that can be performed from home (BQ)	(7) Labor intensity based on number of employees (TQ)	(8) Manually classified as highly affected
Cash / assets × highly affected	0.301**	0.094	0.230**	0.226	−0.014	0.106	−0.159**	0.303**
	(2.53)	(1.11)	(2.58)	(1.50)	(−0.16)	(0.78)	(−2.07)	(2.44)
St debt / assets × highly affected	0.034	0.355	−0.242	−0.268	0.207	−0.343	−0.118	0.139
	(0.14)	(1.18)	(−1.00)	(−1.07)	(0.91)	(−1.46)	(−0.45)	(0.58)
Lt debt / assets × highly affected	0.143**	−0.025	0.156**	0.010	0.004	0.078	−0.178***	0.109*
	(2.36)	(−0.39)	(2.57)	(0.15)	(0.07)	(1.21)	(−2.72)	(1.79)
Cash / assets	0.025	0.037	0.012	0.057	0.058	0.059	0.101**	0.025
	(0.55)	(0.75)	(0.24)	(1.29)	(1.34)	(1.33)	(2.16)	(0.57)
St debt / assets	−0.236*	−0.307**	−0.161	−0.163	−0.245*	−0.104	−0.200	−0.270*
	(−1.67)	(−2.45)	(−1.17)	(−1.21)	(−1.80)	(−0.75)	(−1.54)	(−1.93)
Lt debt / assets	−0.214***	−0.166***	−0.219***	−0.172***	−0.178***	−0.192***	−0.145***	−0.204***
	(−6.17)	(−4.97)	(−6.40)	(−5.32)	(−5.53)	(−5.85)	(−4.58)	(−5.92)
Ind: Highly affected	−0.073**	−0.061	−0.109***	0.038	−0.143***	−0.028	0.068**	−0.096**
	(−2.00)	(−1.51)	(−2.80)	(0.96)	(−4.74)	(−0.74)	(2.15)	(−2.58)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,537	1,537	1,537	1,537	1,528	1,548	1,545	1,537
Adjusted R^2	0.194	0.192	0.196	0.193	0.232	0.191	0.194	0.194

Table 6

Accounting performance (cumulative Q1+Q2), financial flexibility measures interacted with affectedness, and firm characteristics

Cross-sectional regressions of return on assets, defined as cumulative operating income before depreciation for the first two quarters of 2020 divided by total assets at year-end 2019, winsorized at the 1st and 99th percentiles. All regressions include Fama-French 49 industry fixed effects. t-statistics are in parentheses. $^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$.

	(1) ROA	(2) ROA
Cash / assets × highly affected	0.152***	0.051*
	(3.90)	(1.65)
St debt / assets × highly affected	0.106	0.113*
	(1.31)	(1.78)
Lt debt / assets × highly affected	0.019	0.022
	(0.91)	(1.38)
Cash / assets	−0.074***	−0.008
	(−5.73)	(−0.64)
St debt / assets	−0.115**	−0.072*
	(−2.55)	(−1.85)
Lt debt / assets	0.008	−0.007
	(0.73)	(−0.73)
Ind: Highly affected	−0.041***	−0.033***
	(−3.16)	(−3.44)
Payout / assets		0.111***
		(4.18)
Ind: IG debt rating		−0.005
		(−0.88)
Capex / lagged assets		−0.068**
		(−2.01)
SGA / sales		−0.020***
		(−7.79)
COGS / sales		0.001
		(0.07)
RD / assets		−0.123***
		(−4.62)
Firm age		0.000
		(0.23)
Lagged ROA: Q1+Q2 2019 / assets EoFY18	0.570***	0.546***
	(38.62)	(27.17)
Equity beta		−0.005
		(−1.48)
BM at EoY 2019	−0.002	−0.010***
	(−0.50)	(−2.93)
ln(MVE) at EoY 2019	0.006***	0.003**
	(5.76)	(2.47)
Momentum 2019		0.002
		(0.65)
Observations	1,739	1,487
Adjusted R^2	0.731	0.699

Table 7

Payout ratios by quartile and financial flexibility measures

Financial flexibility measures and payout ratios for the entire sample and by payout-ratio quartile. (Cash + payout) / (assets + payout) is the cash-to-assets ratio had the firm retained its payouts; (Lt debt − payout) / assets is the long-term debt ratio had the firm used its payouts to repay long-term debt. The left panel uses 2019 payouts; the right panel uses cumulative 2017–2019 payouts over 2019 assets.

	2019 payouts / 2019 assets			(2017+2018+2019 payouts) / 2019 assets		
	N	Median	Mean	N	Median	Mean
All firms						
Cash / assets	1,843	0.116	0.239	1,816	0.116	0.237
(Cash + payout) / (assets + payout)	1,843	0.155	0.265	1,816	0.215	0.304
Lt debt / assets	1,838	0.244	0.271	1,811	0.245	0.273
(Lt debt - payout) / assets	1,838	0.205	0.234	1,811	0.143	0.158
Payout / assets	1,843	0.011	0.037	1,816	0.038	0.115
Top quartile payout ratio						
Cash / assets	461	0.104	0.160	454	0.107	0.169
(Cash + payout) / (assets + payout)	461	0.182	0.241	454	0.316	0.361
Lt debt / assets	458	0.268	0.302	451	0.265	0.299
(Lt debt - payout) / assets	458	0.175	0.180	451	−0.020	−0.064
Payout / assets	461	0.080	0.122	454	0.251	0.363
Bottom quartile payout ratio						
Cash / assets	460	0.490	0.468	454	0.455	0.454
(Cash + payout) / (assets + payout)	460	0.490	0.468	454	0.455	0.454
Lt debt / assets	460	0.117	0.222	454	0.134	0.233
(Lt debt - payout) / assets	460	0.117	0.222	454	0.133	0.233
Payout / assets	460	0.000	0.000	454	0.000	0.000
Middle quartiles payout ratios						
Cash / assets	922	0.078	0.164	908	0.083	0.163
(Cash + payout) / (assets + payout)	922	0.093	0.176	908	0.133	0.201
Lt debt / assets	920	0.275	0.281	906	0.271	0.279
(Lt debt - payout) / assets	920	0.259	0.268	906	0.216	0.231
Payout / assets	922	0.011	0.014	908	0.038	0.048

Table 8

Stock returns, financial flexibility measures, and a top quartile payout ratio indicator variable

Cross-sectional regressions of excess stock returns on firm characteristics and an indicator equal to one if cumulative 2017–2019 payouts over 2019 assets are in the top quartile. Collapse period: February 3 to March 23, 2020. Stimulus day: March 24, 2020. All regressions include Fama-French 49 industry fixed effects. t-statistics are in parentheses. $^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$.

	(1) Collapse period	(2) Stimulus day	(3) Collapse period	(4) Stimulus day
Ind: 3-year cum. payout / assets top quartile	0.016	−0.009*	0.010	−0.009
	(1.17)	(−1.82)	(0.68)	(−1.64)
Cash / assets			0.086**	0.009
			(1.99)	(0.58)
St debt / assets			−0.251**	0.091**
			(−2.20)	(2.13)
Lt debt / assets			−0.173***	0.030***
			(−5.83)	(2.67)
Ind: IG debt rating			0.017	−0.010
			(0.87)	(−1.38)
Capex / lagged assets			−0.001	0.042
			(−0.01)	(0.86)
SGA / sales			−0.034***	0.009**
			(−3.53)	(2.55)
COGS / sales			0.036	0.003
			(0.97)	(0.20)
RD / assets			0.026	−0.021
			(0.28)	(−0.60)
Equity beta	−0.047***	0.026***	−0.061***	0.025***
	(−4.17)	(6.60)	(−4.83)	(5.26)
BM at EoY 2019	0.019*	−0.011***	0.009	−0.008
	(1.69)	(−2.85)	(0.70)	(−1.62)
ln(MVE) at EoY 2019	0.005*	0.006***	0.005	0.008***
	(1.69)	(5.78)	(1.21)	(5.06)
Momentum 2019	−0.004	0.005	−0.010	0.011**
	(−0.41)	(1.27)	(−0.72)	(2.27)
Profitability	0.010	0.007	0.066*	0.008
	(0.55)	(1.07)	(1.82)	(0.56)
Observations	1,803	1,796	1,538	1,533
Adjusted R^2	0.145	0.126	0.195	0.131

Table 9

Stock returns, financial flexibility measures, and conglomerate indicator variables

Cross-sectional regressions of excess stock returns on firm characteristics and an indicator for COVID-19 conglomerates: firms with segments in two or more NAICS three-digit industries, at least one but not all of which are highly affected. Regressions do not include industry fixed effects. t-statistics are in parentheses. $^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$.

	(1) Collapse period	(2) Stimulus day	(3) Collapse period	(4) Stimulus day
Cash / assets × Ind: COVID-19 conglomerate			0.204	−0.014
			(1.38)	(−0.25)
St debt / assets × Ind: COVID-19 conglomerate			0.904**	−0.082
			(2.38)	(−0.59)
Lt debt / assets × Ind: COVID-19 conglomerate			0.192**	0.030
			(1.96)	(0.84)
Cash / assets × Ind: Highly affected (0 for conglomerates)			0.279**	−0.021
			(2.12)	(−0.43)
St debt / assets × Ind: Highly affected (0 for conglomerates)			0.369	0.001
			(1.49)	(0.01)
Lt debt / assets × Ind: Highly affected (0 for conglomerates)			0.152**	0.065***
			(2.49)	(2.89)
Cash / assets	0.129***	0.006	0.074	0.007
	(3.01)	(0.41)	(1.58)	(0.43)
St debt / assets	−0.088	0.066	−0.289*	0.076
	(−0.77)	(1.59)	(−1.91)	(1.38)
Lt debt / assets	−0.184***	0.046***	−0.241***	0.025*
	(−6.29)	(4.29)	(−6.75)	(1.88)
Ind: COVID-19 conglomerate	−0.009	−0.001	−0.121***	−0.005
	(−0.42)	(−0.12)	(−2.62)	(−0.30)
Ind: Highly affected (set to zero for COVID-19 conglomerate)	−0.057***	0.018***	−0.146***	−0.002
	(−3.66)	(3.15)	(−4.63)	(−0.15)
Ind: IG debt rating	0.000	−0.012*	0.003	−0.011
	(0.01)	(−1.69)	(0.16)	(−1.54)
Capex / lagged assets	−0.312**	0.071	−0.311**	0.063
	(−2.58)	(1.61)	(−2.57)	(1.43)
SGA / sales	−0.015*	0.006*	−0.013	0.006*
	(−1.70)	(1.85)	(−1.38)	(1.81)
COGS / sales	−0.054*	0.032***	−0.061*	0.034***
	(−1.72)	(2.74)	(−1.92)	(2.95)
RD / assets	0.055	−0.016	0.083	−0.020
	(0.59)	(−0.47)	(0.87)	(−0.58)
Payout / assets	−0.060	0.003	−0.059	0.002
	(−0.57)	(0.08)	(−0.56)	(0.04)
Equity beta	−0.072***	0.022***	−0.068***	0.023***
	(−6.08)	(5.11)	(−5.77)	(5.29)
BM at EoY 2019	0.001	−0.004	0.004	−0.004
	(0.07)	(−0.85)	(0.31)	(−0.89)
ln(MVE) at EoY 2019	0.009**	0.008***	0.009**	0.007***
	(2.09)	(4.85)	(2.09)	(4.73)
Momentum 2019	−0.008	0.018***	−0.009	0.018***
	(−0.62)	(3.58)	(−0.70)	(3.60)
Profitability	0.113***	0.002	0.111***	0.003
	(3.24)	(0.16)	(3.17)	(0.25)
One-sided p: conglomerate > highly affected	0.016	0.989	0.311	0.572
Industry fixed effects	No	No	No	No
Observations	1,537	1,532	1,537	1,532
Adjusted R^2	0.124	0.093	0.131	0.095

Table 10

Stock returns and measures of financial constraints

Cross-sectional regressions of excess stock returns on indicators of financial constraints. Collapse period: February 3 to March 23, 2020. Stimulus day: March 24, 2020. All regressions include the asset pricing characteristics and Fama-French 49 industry fixed effects. t-statistics are in parentheses. $^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$.

	(1) Collapse	(2) Stimulus	(3) Collapse	(4) Stimulus	(5) Collapse	(6) Stimulus	(7) Collapse	(8) Stimulus
FC (WW index)	0.051**	−0.006						
	(2.50)	(−0.81)						
FC (KZ index)			−0.035***	0.016***				
			(−2.64)	(3.28)				
FC (SA index)					0.010	0.001		
					(0.63)	(0.21)		
FC (HR cash ex ante)							−0.028**	−0.004
							(−2.00)	(−0.79)
Equity beta	−0.053***	0.030***	−0.046***	0.024***	−0.046***	0.027***	−0.047***	0.026***
	(−4.64)	(7.19)	(−4.17)	(6.11)	(−4.31)	(6.94)	(−4.41)	(6.74)
BM at EoY 2019	0.018	−0.010**	0.022*	−0.012***	0.018*	−0.009**	0.016	−0.009**
	(1.56)	(−2.49)	(1.95)	(−2.89)	(1.65)	(−2.16)	(1.51)	(−2.26)
ln(MVE) at EoY 2019	0.009***	0.005***	0.006*	0.006***	0.007**	0.006***	0.006*	0.006***
	(2.64)	(4.33)	(1.82)	(5.61)	(2.18)	(5.16)	(1.88)	(5.65)
Momentum 2019	0.003	0.005	−0.001	0.006*	−0.005	0.007*	−0.006	0.006*
	(0.23)	(1.31)	(−0.09)	(1.71)	(−0.53)	(1.78)	(−0.61)	(1.69)
Profitability	0.011	0.006	0.007	0.000	0.012	0.006	0.007	0.005
	(0.64)	(0.89)	(0.36)	(0.02)	(0.70)	(0.95)	(0.39)	(0.79)
Observations	1,722	1,715	1,789	1,782	1,828	1,821	1,824	1,817
Adjusted R^2	0.154	0.119	0.148	0.123	0.145	0.122	0.146	0.122

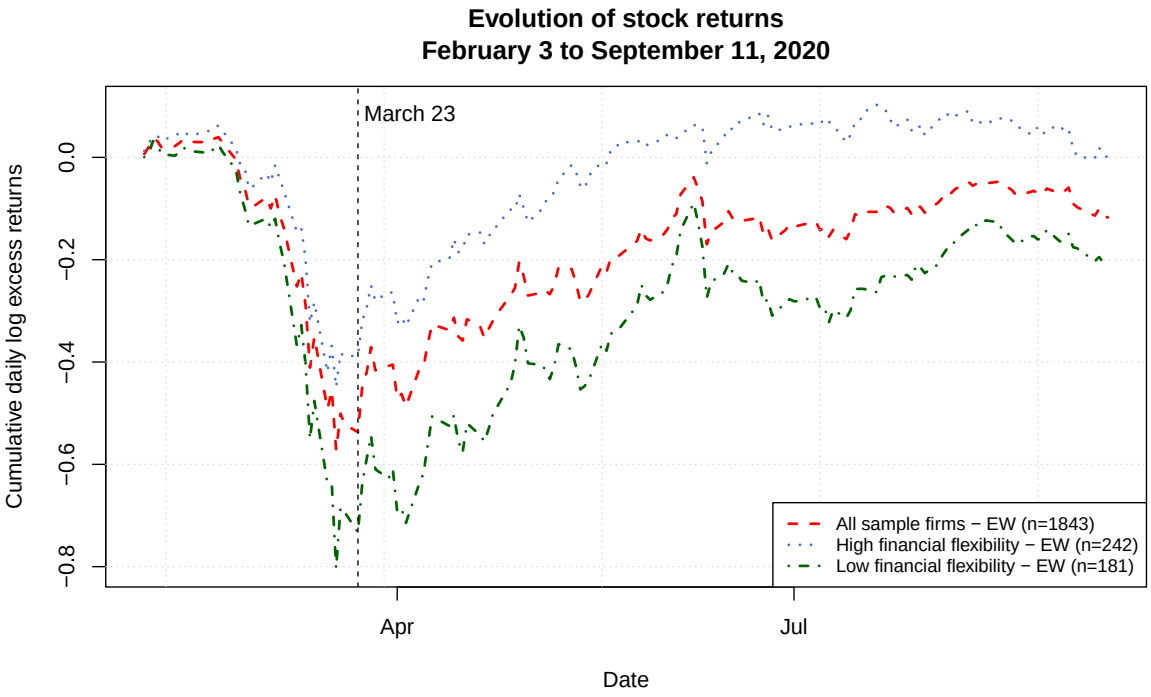


Figure 1

Evolution of stock returns for groups of firms with high and low financial flexibility

Cumulative daily log excess stock returns from February 3 to September 11, 2020, for an equal-weighted portfolio of all sample firms, a portfolio of firms with high financial flexibility (top quartile of cash over assets and bottom quartile of long-term debt over assets at the end of fiscal year 2019), and a portfolio of firms with low financial flexibility (bottom quartile of cash and top quartile of long-term debt).

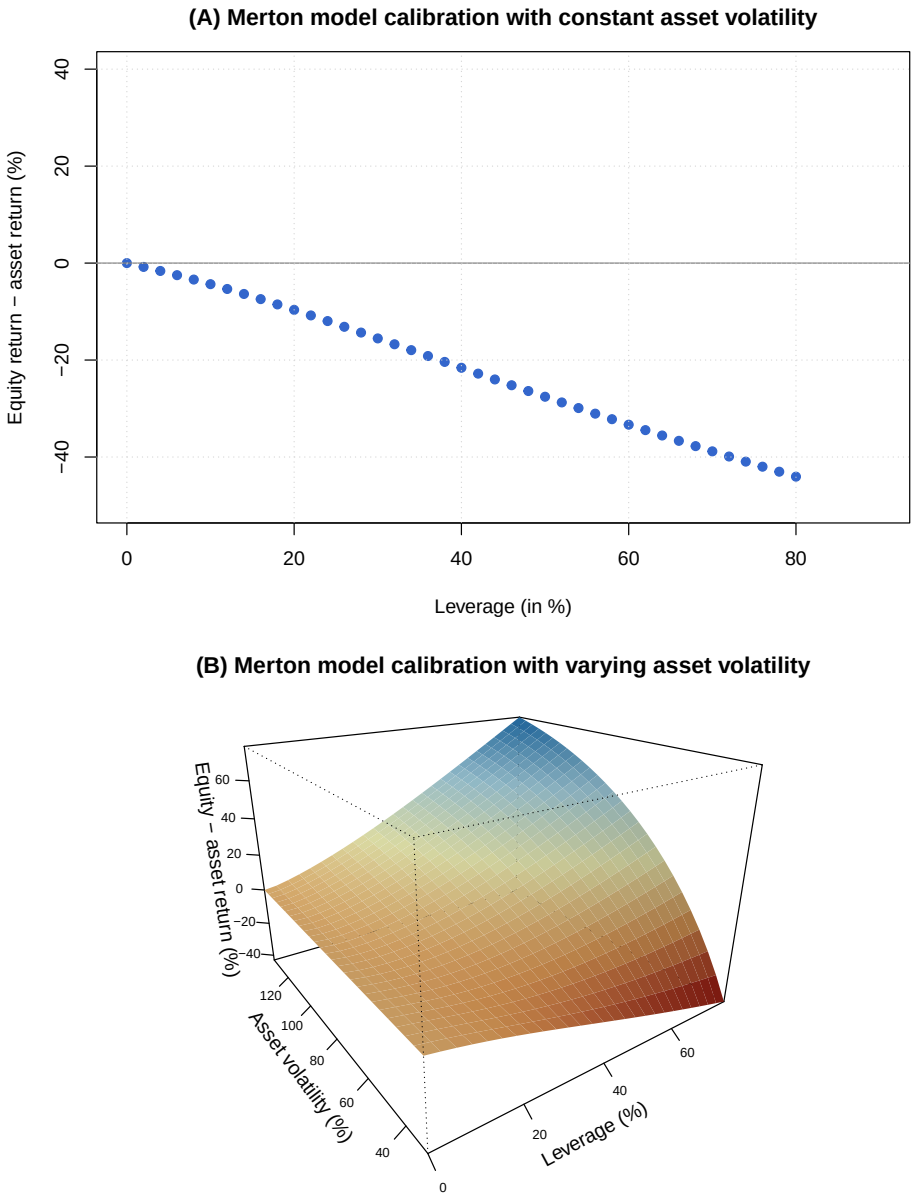


Figure 2
Calibration of the Merton model

Panel A shows the difference between equity and asset returns for a large shock to asset value as a function of leverage (the face value of debt) in a calibration of the Merton (1974) model, with $V_0 = 100$, $V_{Shock} = 70$, $\sigma_{assets} = 30\%$, $r = 3\%$, $q = 1\%$, and $T = 5$ years. Panel B shows the same difference as a function of both leverage and asset volatility.